

Prototype Parts List & Node BOM

Everything you need to build a working gunshot-sensor node on your bench, then scale to the five field node types. Every part links to a vendor page. The brain + software are built and proven (97.8% recall on 8,015 real gunshots); this is the shopping list to put it on hardware.

01 Start here — fastest bench prototype (~\$85)

Get one real Pi hearing a real shot into the C2. No enclosure, no weatherproofing — those come after you see it work. Pick ONE mic (the SPH0645 is recommended; a USB dongle is the zero-driver fallback).

Part	Purpose	Price	Link
Raspberry Pi 4 (2 GB)	The brain; runs the exact detector unchanged	~ \$45	Buy »
Official Pi 4 USB-C PSU (5V/3A)	Clean bench power	~ \$8	Buy »
Industrial microSD 32 GB	24/7 endurance; pair with read-only rootfs	~ \$12	Buy »
Adafruit SPH0645 I2S mic [recommended]	Standard I2S, 120 dB AOP, huge Pi docs	• \$6.95	Buy »
- or Adafruit ICS-43434 I2S mic	Cleaner data, no DC-offset quirk (chip EOL)	• \$8.95	Buy »
- or USB audio dongle (CM108)	Zero drivers - most robust bring-up	~ \$10	Buy »
Jumper wires / breadboard	Wire the I2S mic to the GPIO header	~ \$5	Buy »

Wiring (SPH0645 -> Pi 4): 3V->3V3 · GND->GND · BCLCK->GPIO18 · LRCLK->GPIO19 · DOUT->GPIO20. Add `dtparam=i2s=on + dtoverlay=googlevoicehat-soundcard` to `/boot/firmware/config.txt`, reboot, verify with `arecord -l`.

02 The five node types

A site is mostly cheap leaves hopping to a gateway. Unit costs are materials incl. housing.

Node type	Brain	Uplink	Power	Housing	Unit \$
Indoor leaf	Pi Zero 2 W	LoRa -> gw	building + battery	ceiling grille, tamper	~\$150
Indoor gateway	Pi 4	LoRa + cellular	building + battery	ceiling grille (larger)	~\$230
Roof gateway	Pi 4	LoRa + cellular	building + battery	Hammond IP68 + Gore vent	~\$270
Vehicle	Pi 4	LoRa + cellular	12V + battery	trunk box + roof mic puck	~\$290
Drone	Zero 2 W / ESP32	LoRa (+cell)	drone / LiPo	ultralight shell + clip	~\$55

03 Field parts catalog

The parts that turn a bench Pi into fielded leaves, gateways, roof, vehicle, and drone nodes. Confirm the ~ prices before ordering (some are street/snippet; • = confirmed from a vendor page).

Mesh & power (all fixed nodes)

Part	Purpose	Price	Link
LoRa radio (SX1262 HAT)	Mesh transport - leaves hop to a gateway	~ \$15-20	Buy »

Part	Purpose	Price	Link
Battery UPS HAT + cells (LiFePO4 for hot ceilings)	Rides through a power outage; auto-recharge	~ \$40-70	Buy »
Mains USB-C adapter or PoE splitter	Building power to charge the battery	~ \$8-20	Buy »
Pi Zero 2 W (cost-down leaf / drone brain)	~1 W featherweight brain	~ \$17	Buy »

Gateway uplink (1+ per site; every cruiser)

Part	Purpose	Price	Link
Cellular HAT (Waveshare SIM7080G, LTE-M)	Uplink independent of building network	~ \$40-50	Buy »
Hologram SIM (alert-only data plan)	Tiny JSON alerts; ~\$1-2/mo/node	~ \$1-2/mo	Buy »

Roof / outdoor housing (solved)

Part	Purpose	Price	Link
Hammond 1554T2GY enclosure	UV-poly, IP68/NEMA 4X, RF-transparent	• \$41.78	Buy »
Gore GAW334 acoustic vent (x4)	The mic port - ePTFE, IP68, ~1.4 dB loss	• \$10/4pc	Buy »
Amphenol pressure vent (VENT-PQ1NBK)	Pressure equalization; a sealed box needs it	• \$2.39	Buy »
Hammond 1427BCG brass cable gland	IP68 - brass, nylon photodegrades outdoors	~ \$4	Buy »

Vehicle node (split: trunk brain + roof mic/antennas)

Part	Purpose	Price	Link
Pololu D24V50F5 buck (6-38V -> 5V/5A)	Automotive-grade wide input	• \$29.95	Buy »
Automotive TVS diode (SMBJ) + inline fuse	Clamp load-dump spikes past the buck	~ \$5	Buy »
Supercap UPS HAT (graceful shutdown)	Clean shutdown on ignition-off	~ \$25-35	Buy »
GPS module (u-blox MAX-M10S)	Position + speed gate + TDOA time	~ \$40	Buy »
NMO antenna mount (Larsen MB8X)	Fleet-accepted roof penetration	• \$39	Buy »

• price confirmed from a vendor/datasheet page | ~ street/estimate - verify before ordering | prices mid-2026, materials only.
 Full model + housing detail: docs/SENTINEL-HARDWARE-BOM.md · architecture: docs/SENTINEL-SHOTS-FIRED-ARCH.md · county case study: docs/SENTINEL-ESCAMBIA-CASE-STUDY.md. The high-AOP mic (Infineon IM73A135) price was unverifiable at research time - confirm live.